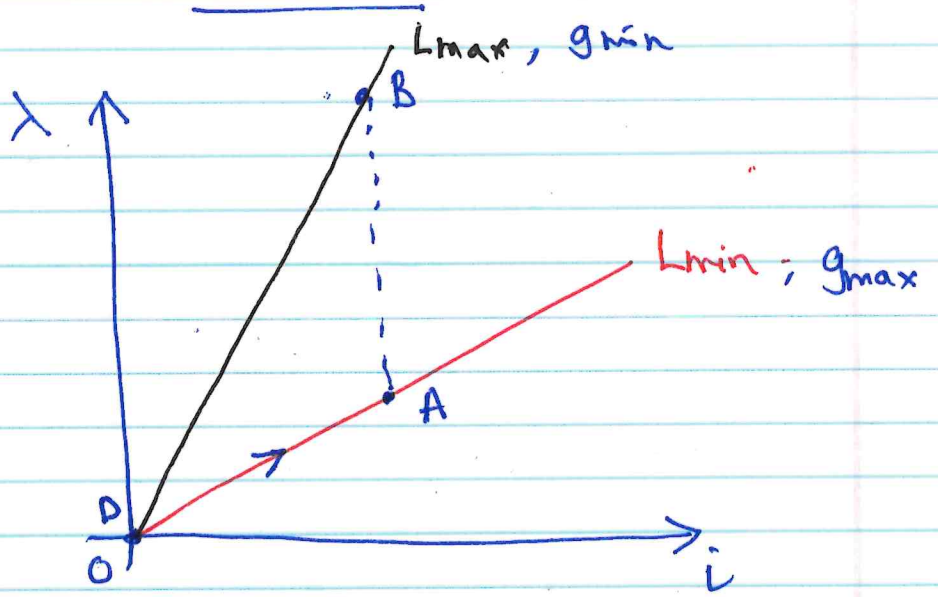
Lecture 19

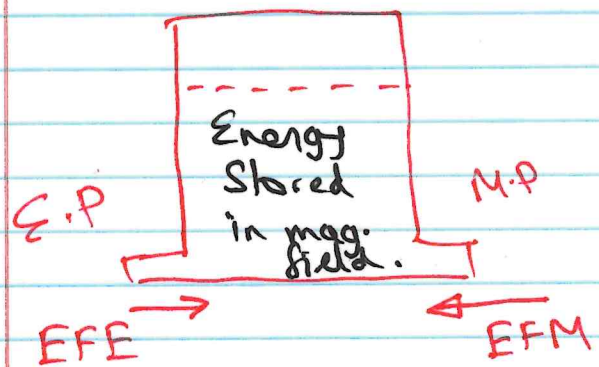


O → A → B → D : Motor / Generator

↓

net energy

$\mathcal{E}$ port → M·port



net energy

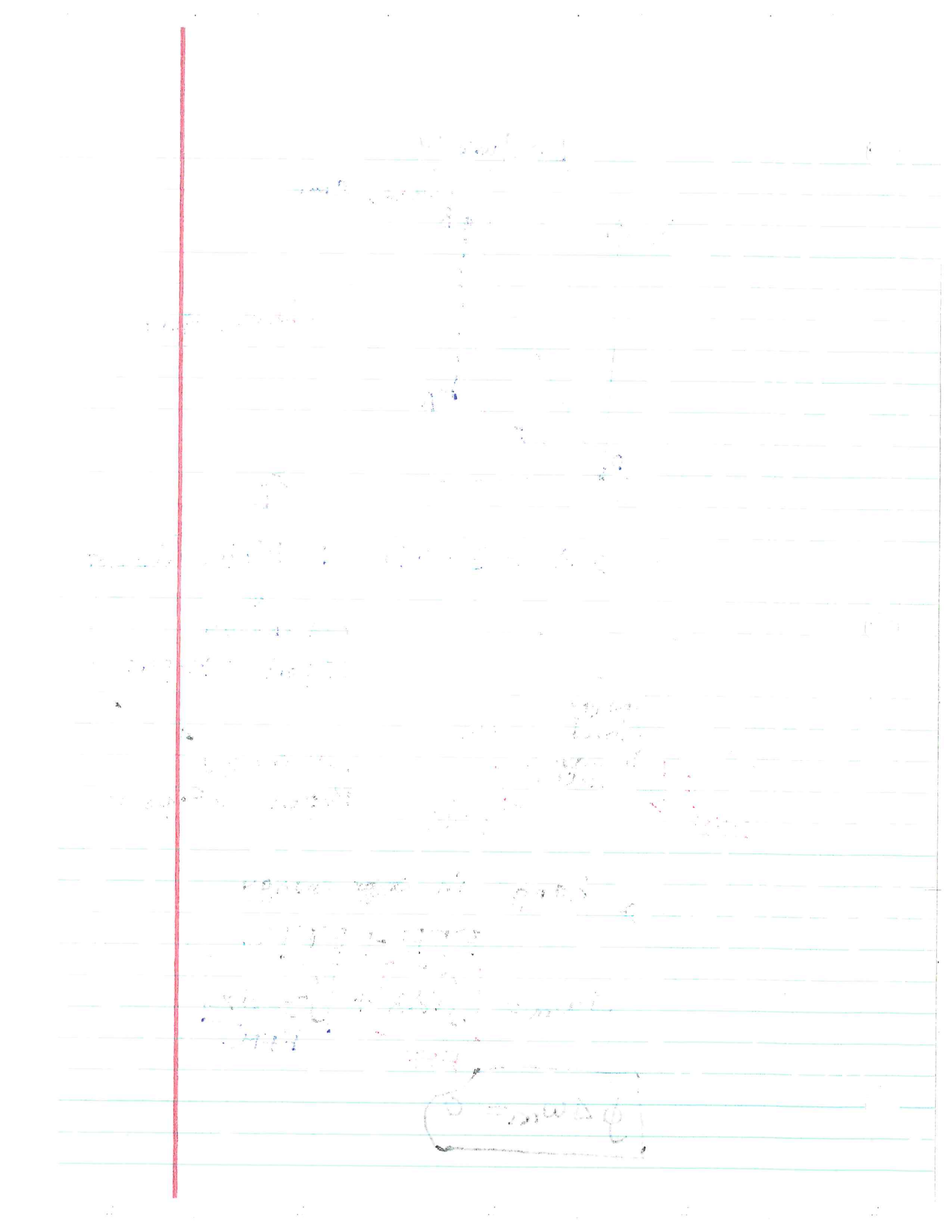
M·port → $\mathcal{E}$·port

Change in ~~energy~~ energy

= EFE + EFM.

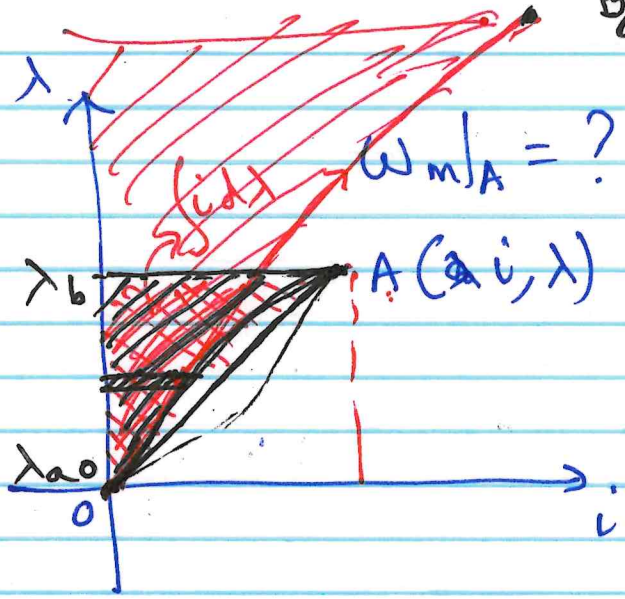
$$\Delta\omega_m = \boxed{\int i\,d\lambda}_{\text{EFE}} + \boxed{\int -f_e\,dx}_{\text{EFM}}$$

$$\boxed{\oint \Delta\omega_m = 0}$$



$\omega_m|_B$

Detour :



$$EFE = \int_{\lambda_a}^{\lambda_b} i\,d\lambda = \omega_m|_A$$

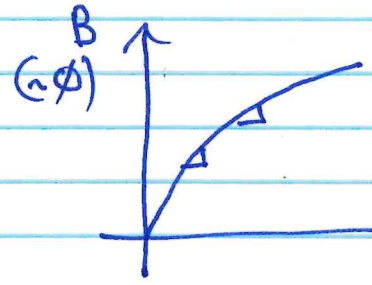
$$d\omega_m = i\,d\lambda - f_x\,dx$$

$x = \text{const.}$

$$\lambda = f(x, i)$$

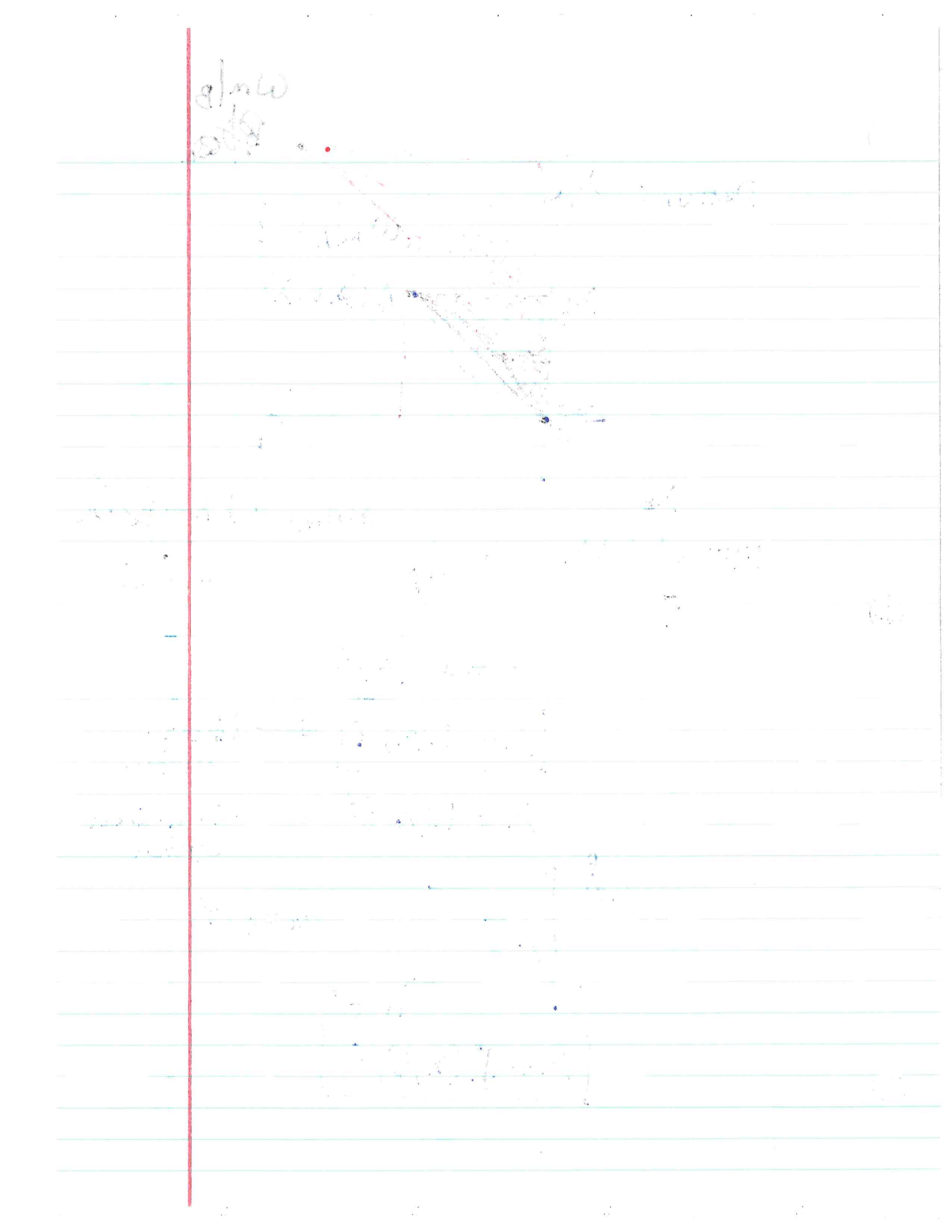
$$\boxed{\lambda = L(x)\, i} \leftarrow \text{Linear system}$$

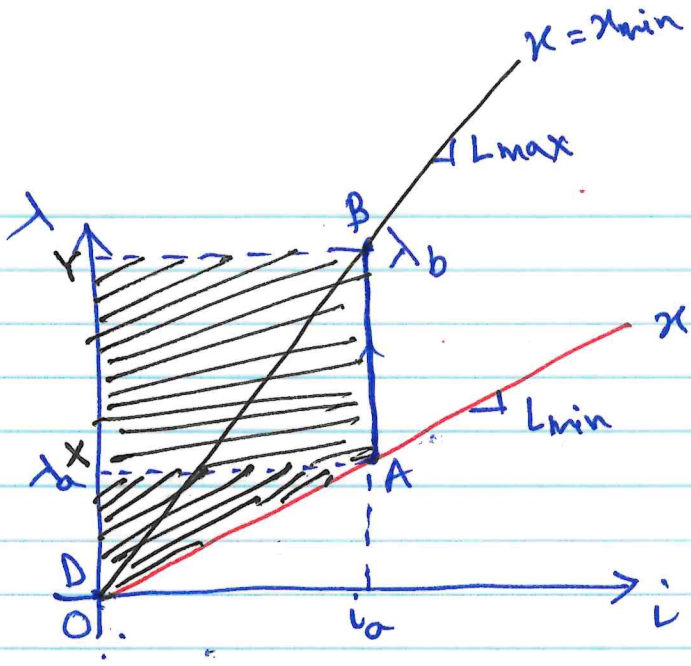
$$\lambda = L(x)\, i^3 \leftarrow \text{Non-linear system}$$



$$\mu_r = f(I)$$

$$\boxed{\lambda = L(x, i)\, i}$$

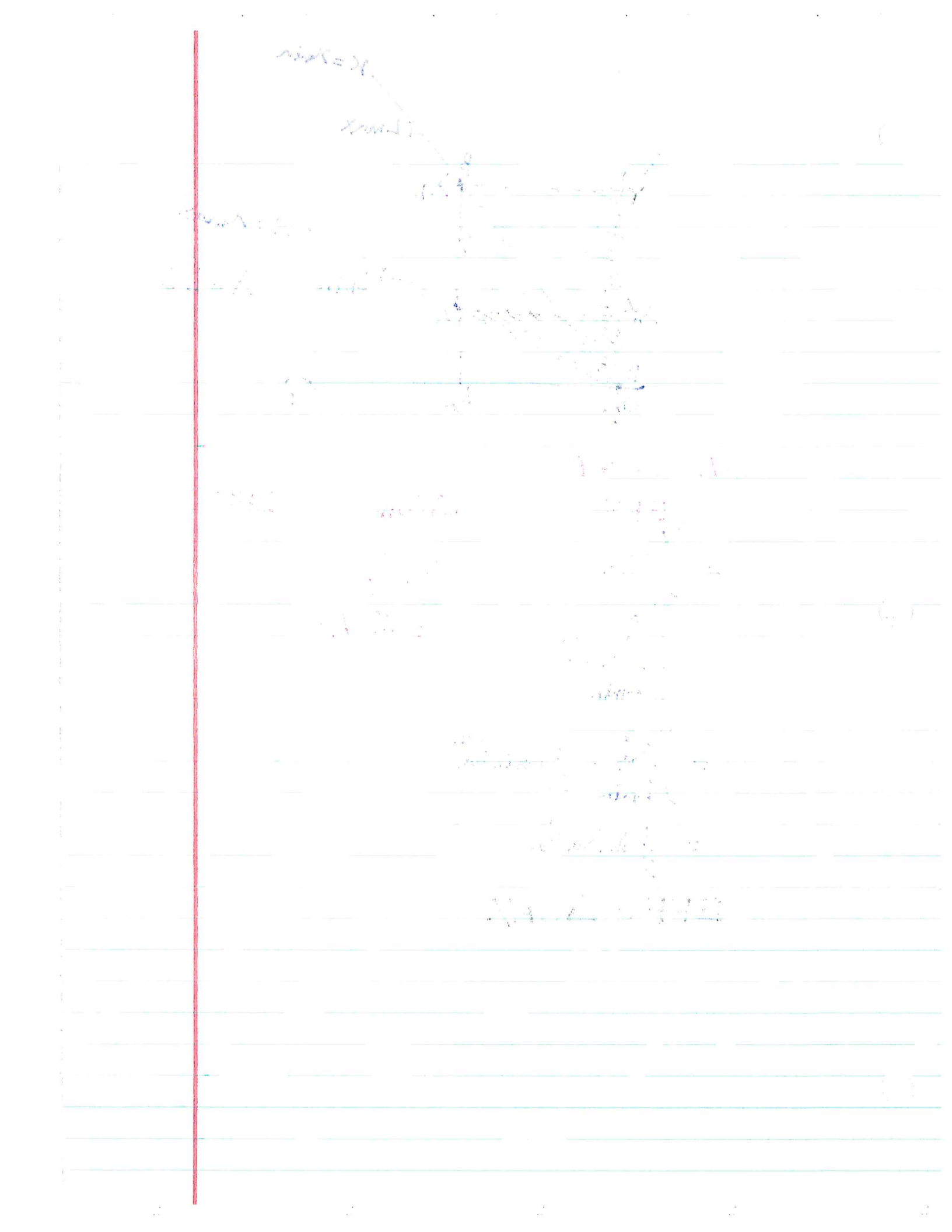




$O \rightarrow A$

EFE	$\Delta\omega_m$	EFM
$= \int_0^{\lambda_a} i\,d\lambda$	$\omega_m\|_A$	0
$= \int_0^{\lambda_a} \frac{\lambda}{L_{min}}\,d\lambda$	$= \Delta OAX$	
$= \frac{\lambda_a^2}{2L_{min}} = \frac{1}{2} L_{min} i_a^2$		
$= \frac{1}{2} \lambda_a i_a$		

$EFE = \Delta OAX$



$A \rightarrow B$

EFE $= \int_{\lambda_a}^{\lambda_b} i \, d\lambda$

$= i_a(\lambda_b - \lambda_a)$

$= \square XABY$

$\Delta \omega_m$

$\omega_m|_B - \omega_m|_A$

$= \Delta OBY - \Delta OAX$

x_b EFM.

$= -\int_{x_a}^{x_b} f_e \, dx$

EFM $= \Delta OBY - \Delta OAX$

$- \square XABY$

$B \rightarrow D$

EFE

$= \int i \, d\lambda$

$= -\Delta OBY$

$\Delta \omega_m = -\Delta OBY$

$\omega_m|_D = 0$

EFM

0

$D \rightarrow 0$

EFE $= 0$

$\Delta \omega_m = 0$

$\omega_m|_0 = 0$

EFM

0

$$0 = \oint \Delta\omega_m = \oint EFE + \oint EFM$$

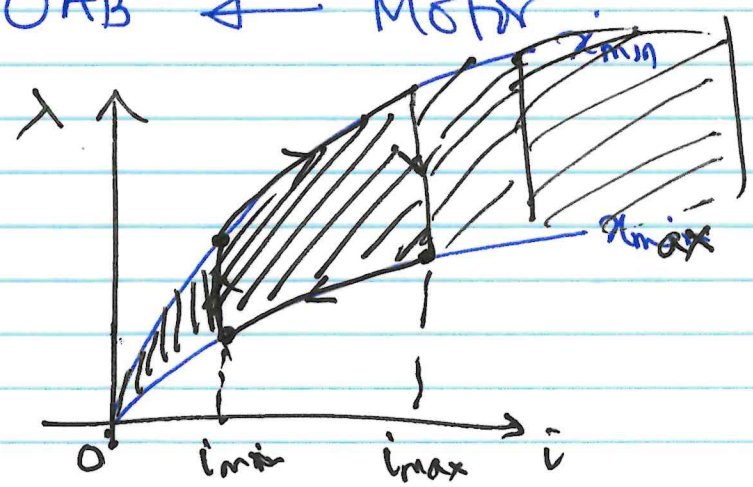
$$\boxed{\oint EFE + \oint EFM = 0}$$

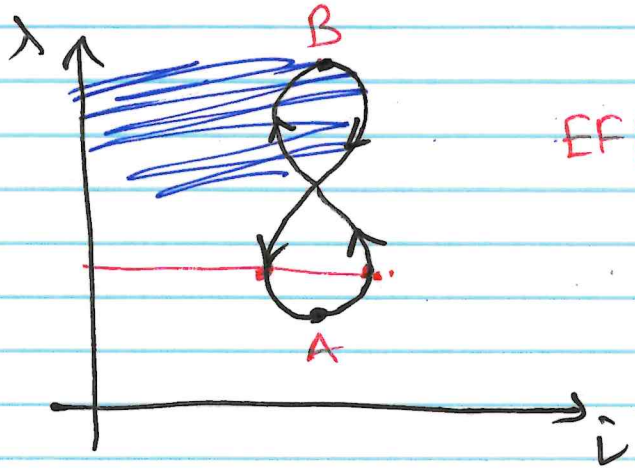
$$\oint_{A} EFE = -\oint_{B} EFM > 0$$

$$\int_{A}^{B} EFE + \int_{B}^{A} EFE + \int_{B}^{D} EFE + \int_{D}^{O} EFE$$

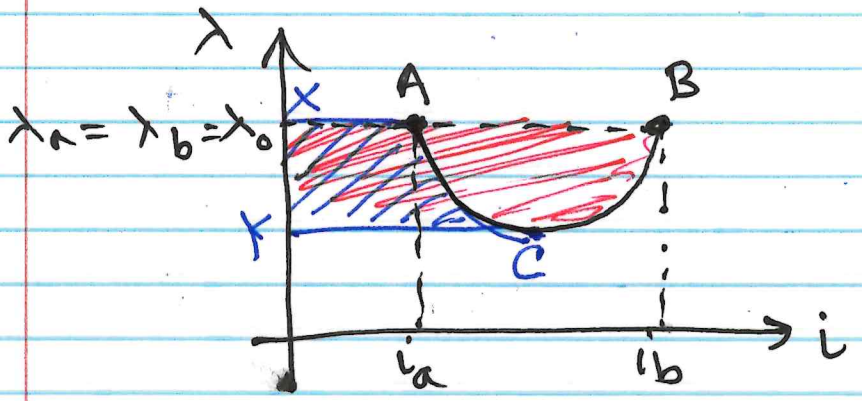
$$= \Delta OAX + \square XABY - \Delta OBY + 0$$

$$= \Delta OAB \leftarrow \text{Motor}$$





EFE = A → B



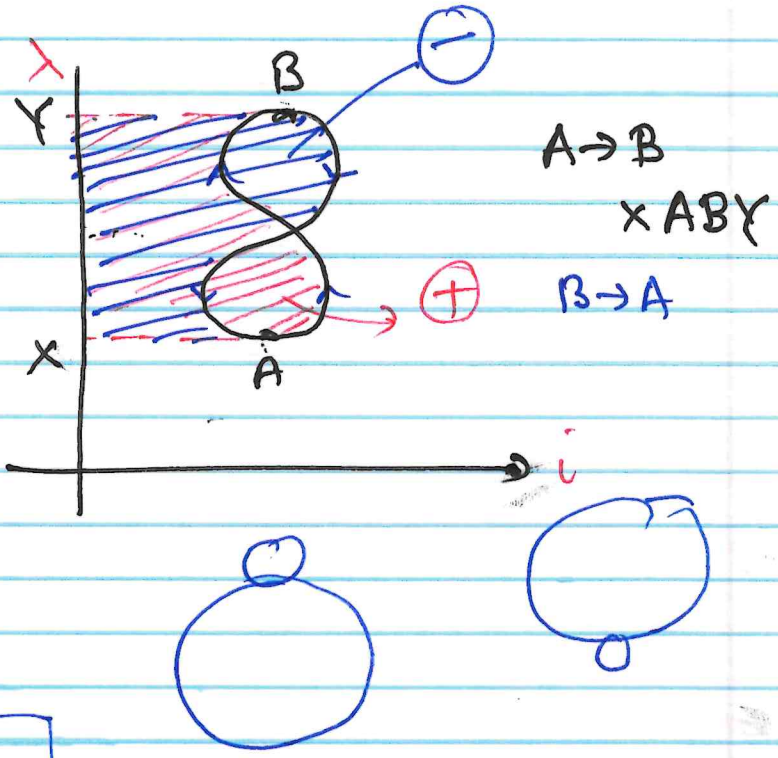
EFE = ?

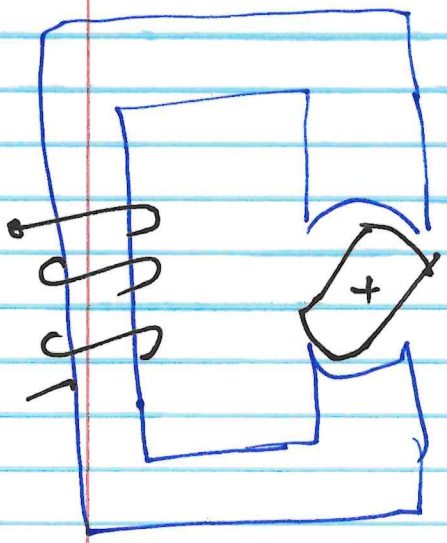
$$EFE = \int_{\lambda_a}^{\lambda_b} i\,d\lambda \neq 0$$

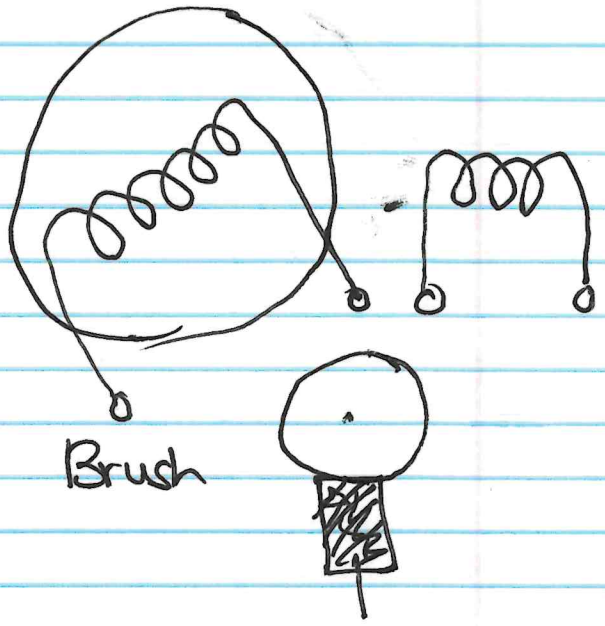


A → C EFE = −XACY

C → B EFE = +XBCY


Y
B
A
X
i
A→B
xABY
B→A


+


Brush



